1 Write down the equation of a line parallel to $y = 3x + 2$

[$y = 3x \pm$ Anything]

$y = 3x + 1$

(Total for question 1 is 1 mark)

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2 Write down the equation of the line parallel to $y = \frac{1}{2}x + 5$ which passes through (0,2)

$y = \frac{1}{2}x + 2$

(Total for question 2 is 2 marks)

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3 Write down the equation of the line parallel to $y = -x + 1$ which passes through (0,–4)

$y = -x - 4$

(Total for question 3 is 2 marks)

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4 Write down the equation of a line perpendicular to $y = 3x + 3$

$m = -\frac{1}{3}$

[$y = -\frac{1}{3}x \pm$ Anything]

$y = -\frac{1}{3}x + 1$

(Total for question 4 is 1 mark)

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5 Write down the equation of the line perpendicular to $y = \frac{1}{2}x - 4$ which passes through (0,7)

$m = -2$

$y = -2x + 7$

(Total for question 5 is 2 marks)

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6 Write down the equation of the line perpendicular to $y = -\frac{3}{2}x - 1$ which passes through (0,–8)

$m = \frac{2}{3}$

$y = \frac{2}{3}x - 8$

(Total for question 6 is 2 marka)

7 Find the equation of the line parallel to $2y - 3x + 2 = 0$ which passes through (0,4)

$2y = 3x - 2$

$y = \frac{3}{2}x - 1$

$m = \frac{3}{2}$

$y = \frac{3}{2}x + 4$

(Total for question 7 is 2 marks)

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8 Find the equation of the line parallel to $2x + 5y = 10$ which passes through (0,–3)

$5y = -2x + 10$

$y = \frac{-2}{5}x + 2$

$m = -\frac{2}{5}$

$y = -\frac{2}{5}x - 3$

(Total for question 8 is 2 marks)

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9 Find the equation of the line perpendicular to $5y = 2x - 4$ which passes through (0,7)

$y = \frac{2}{5}x - \frac{4}{5}$

perpendicular $m = -\frac{5}{2}$

$y = -\frac{5}{2}x + 7$

(Total for question 9 is 2 marks)

10 Here are the equations of five straight lines.

Line A $y = 2x - 3$
Line B $2y = x + 3$
Line C $4y = 3x - 2$
Line D $2y = 4x - 1$
Line E $3y = 2x - 2$

A: $m = 2$
B: $y = \frac{1}{2}x + \frac{3}{2}$ $m = \frac{1}{2}$
C: $y = \frac{3}{4}x - \frac{1}{2}$ $m = \frac{3}{4}$
D = $y = 2x - \frac{1}{2}$ $m = 2$

Two of these lines are parallel.

Write down the two parallel lines.

Line ...A..... and Line ...D.....

(Total for question 10 is 1 mark)

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11 Here are the equations of five straight lines.

Line A $y + 3x = 4$
Line B $2y = x + 1$
Line C $y + 2x = 3$
Line D $y = 4x - 2$
Line E $2y = 2x - 1$

A: $y = -3x + 4$ $m = -3$
B: $y = \frac{1}{2}x + \frac{1}{2}$ $m = \frac{1}{2}$
C: $y = -2x + 3$ $m = -2$

Two of these lines are perpendicular.

Write down the two perpendicular lines.

Line ...B..... and Line ...C.....

(Total for question 11 is 1 mark)

$x_1 \; y_1 \quad x_2 \; y_2$

12 Line A passes through the points (2, 1) and (5, 10)
Find the equation of the line parallel to A that passes through (2,5)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{10 - 1}{5 - 2} = \frac{9}{3} = \underline{\underline{3}}$$

$y = 3x + c \qquad (2, 5)$ $x \; y$

$5 = 3(2) + c$

$5 = 6 + c$

$\underline{\underline{c = -1}}$

$y = 3x - 1$

(Total for question 12 is 3 marks)

$x_1 \; y_1 \quad x_2 \; y_2$

13 Line A passes through the points (1, 5) and (5, 7)
Find the equation of the line perpendicular to A that passes through (-1,7)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{7 - 5}{5 - 1} = \frac{2}{4} = \frac{1}{2}$$

perp. $m = -2$

$y = -2x + c \qquad (-1, 7)$

$7 = -2(-1) + c$

$7 = 2 + c$

$\underline{\underline{c = 5}}$

$y = -2x + 5$

(Total for question 13 is 2 marks)

$x_1 \; y_1 \quad x_2 \; y_2$

14 Line A passes through the points (-2, 1) and (4, 10)
Find the equation of the line parallel to A that passes through (2,7)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{10 - 1}{4 - -2} = \frac{9}{6} = \frac{3}{2}$$

$y = \frac{3}{2}x + c \qquad (2, 7)$

$7 = \frac{3}{2}(2) + c$

$7 = 3 + c$

$\underline{\underline{c = 4}}$

$y = \frac{3}{2}x + 4$

(Total for question 14 is 3 marks)

$x_1 \; y_1 \quad x_2 \; y_2$

15 Line A passes through the points (2, -5) and (10, -1)
Find the equation of the line perpendicular to A that passes through (4,3)

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{-1 - -5}{10 - 2} = \frac{4}{8} = \frac{1}{2}$$

perp $m = -2$

$y = -2x + c \qquad (4, 3)$

$3 = -2(4) + c$

$3 = -8 + c$

$c = 11$

$y = -2x + 11$

(Total for question 15 is 2 marks)

$x_1 \; y_1 \quad x_2 \; y_2$

16 Line A passes through the points (2, 1) and (5, 10)
Line B passes through the points (4, 7) and (2, 1)
Show that Line A and Line B are parallel.

A: $m = \dfrac{y_2 - y_1}{x_2 - x_1}$

$= \dfrac{10-1}{5-2} = \dfrac{9}{3} = 3$

$\underline{\underline{m = 3}}$

B: $m = \dfrac{1-7}{2-4} = \dfrac{-6}{-2} = 3$

$\underline{\underline{m = 3}}$

Same gradient ∴ parallel.

(Total for question 16 is 4 marks)

$x_1 \; y_1 \quad x_2 \; y_2$

17 Line A passes through the points (1, 5) and (5, 7)
Line B passes through the points (-1, 7) and (2, 1)
Show that Line A and Line B are perpendicular.

A: $m = \dfrac{7-5}{5-1} = \dfrac{2}{4} = \underline{\underline{\dfrac{1}{2}}}$

B: $m = \dfrac{1-7}{2--1} = \dfrac{-6}{3} = \underline{\underline{-2}}$

$\dfrac{1}{2} \times -2 = -1$ ∴ A and B are perpendicular.

(Total for question 17 is 4 marks)

$x_1 \; y_1 \quad x_2 \; y_2$

18 Line A passes through the points (3, 6) and (5, -2)
Line B passes through the points (2, 5) and (8, k)

Line A and Line B are parallel.

Find the value of k.

A: $m = \frac{-2-6}{5-3} = \frac{-8}{2} = -4$

B's gradient is -4

$-4 = \frac{k-5}{8-2}$

$-4 = \frac{k-5}{6}$

$-24 = k - 5$

$-19 = k$

$k =$ ....−19....

(Total for question 18 is 4 marks)

$x_1 \; y_1 \quad x_2 \; y_2$

19 Line A passes through the points (-3, -1) and (-1, 9)
Line B passes through the points (-2, 1) and (k, 4)

Line A and Line B are perpendicular.

Find the value of k.

A: $m = \frac{9--1}{-1--3} = \frac{10}{2} = 5$

perp $m = -\frac{1}{5}$

B: $-\frac{1}{5} = \frac{4-1}{k--2}$

$\frac{-1}{5} = \frac{3}{k+2}$

$-(k+2) = 15$

$-k - 2 = 15$

$-2 = 15 + k$

$k = -17$

$k =$ ....−17....

(Total for question 19 is 4 marks)